

Geometry-Aware Sharpness Minimization: Invariance and Generalization in SAM Variants

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1 Motivation

Recent advances in deep learning optimization highlight the importance of sharpness-aware methods that seek flat minima to improve generalization. One such method, Sharpness-Aware Minimization (SAM) [Foret et al., 2021], has shown promising results. However, it is sensitive to the choice of model parameterization, as its perturbation radius is defined in the Euclidean metric.

Adaptive SAM (ASAM) [Kwon et al., 2021] and Monge SAM (M-SAM) [Jacobsen and Arvanitidis, 2025] try to address these limitations by introducing scale invariance and geometry-aware metrics, respectively. However, existing evaluations of M-SAM are limited to a single fine-tuning task on ResNet-18 and a multi-modal alignment task; a systematic comparison of all three optimizers under controlled reparametrization across multiple architectures has not been done.

Understanding how these methods behave under reparametrization is crucial for developing more robust optimization techniques, and this gap motivates our empirical work on benchmarking SAM, ASAM, and M-SAM on ResNet-18 and ViT-B/16 architectures with a focus on reparametrization invariance and generalization performance.

2 Methods

Starting with CIFAR-10, we will train ResNet-18 and ViT-B/16 models using SGD, SAM, ASAM, and M-SAM optimizers. We will track the following metrics to establish a baseline: test (generalization) accuracy, training accuracy, NLL loss curves over epochs, divergence rate across perturbation radii ρ , and Standard Error of the Mean (SEM) across multiple random seeds. Specific values of hyperparameters (batch size, learning rate, number of epochs, perturbation radius ρ , etc.) will be determined during implementation.

Function-preserving reparametrizations (e.g., scaling weights in a ReLU layer by α and compensating in the next layer by $1/\alpha$) will be applied to the network initialization prior to training, to evaluate if geometry-aware methods like M-SAM lead to improved invariance and generalization.

3 Intended Experiments

We will conduct the following experiments:

- **Baseline comparison:** Train ResNet-18 and ViT-B/16 using SGD, SAM, ASAM, and M-SAM on CIFAR-10 and compare the following metrics across optimizers: test accuracy, training accuracy, NLL loss curves, and divergence rate across a sweep of perturbation radii ρ . SGD serves as the reference to confirm that sharpness-aware optimizers provide measurable benefit. Experiments will be run with multiple random seeds (to be fixed during implementation) and results reported with Standard Error of the Mean (SEM).
- **Reparametrization invariance test:** Function-preserving transformations (scaling weights in a ReLU layer by $\alpha \in \{0.1, 0.5, 1, 2, 10\}$ and compensating in the next layer by $1/\alpha$) will be applied to the network initialization prior to training. For each optimizer, we measure the variance of final test accuracy across the five α values. We hypothesize that SAM will be sensitive to reparametrization, ASAM will be partially invariant, and M-SAM will be the most invariant.
- **Flatness analysis:** Visualize loss landscapes along random directions and measure sharpness via $\text{tr}(H)/d$ estimated using the Hutchinson estimator (stochastic trace estimation with Rademacher vectors), which provides a scalable measure of average curvature without forming the full Hessian. This will be used to quantitatively compare the flatness of minima found by each optimizer.

- **(Stretch goal)** Metric ablation (for M-SAM): Replace the Monge metric with alternative metrics such as the Fisher Information Matrix and compare generalization performance versus computational cost.

4 Team Contributions

- **Team Member 1:** Manav Dahra - End-to-end implementation of SGD, SAM, ASAM, and M-SAM; conducting experiments on ResNet-18 and ViT-B/16; analyzing results and writing the final report.

References

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Albert Kjøller Jacobsen and Georgios Arvanitidis. Monge SAM: Robust reparameterization-invariant sharpness-aware minimization based on loss geometry. *arXiv preprint arXiv:2502.08448*, 2025.